

Contextual Ontology Extraction and Inspection of Temporal Narrative Networks

Supplementary Material

1. Reading the networks

The figures show which relationships the model extracted, how they relate to a reader’s question, and where errors remain. They use the same outputs, reference answers and assessments reported in the main paper. No additional experiments were performed.

The two approaches.

- **Pre-extract/select (A)** first extracts relationships without seeing the question. Fixed filters then select records for that question.
- **Contextual extraction (B)** receives the text and question together and generates its own relationships.

A and B are alternative approaches. B does not modify or continue A’s graph.

Reading an arrow. Each arrow runs from the subject to the object. Its label describes their relationship and identifies the supporting text. Labels also show any recorded time interval or attribution, such as whose belief is represented. A believer label describes the existing relationship and does not add another extracted link. In the figures, underscores appear as spaces and `loc` means `located_in`.

Within each panel, exactly matching names share a node. Different names remain separate, even when they might refer to the same entity. Multiple records connecting the same nodes remain separate arrows. Omitted time or attribution fields are `null`, meaning unspecified. A missing time interval does not mean that a relationship holds forever.

Understanding the assessment labels. Table S1 explains the assessment symbols.

Table S1: Assessment symbols used in the network figures.

+	Meaning supported by the text
~	Meaning partly supported
×	Meaning unsupported
?	Interpretation unresolved
I	Supported, but outside the question’s scope

Symbols and line styles accompany the colours, so colour alone is not needed to read the assessments.

In Orchard, node colours distinguish people, objects and places. These categories were added for display and were not inferred by the model. The broad extraction shown in the main paper has no assessment labels. It displays the model’s output rather than a verified reference answer.

Compact index of retained examples

Table S2 indexes the retained examples.

Call numbers are local to their batch. Three pre-extractions precede contextual requests in each batch. A pre-extraction reused for several questions is still one model call. The main paper contains the complete aggregate tables. This supplement retains selected explanations instead of repeating every prompt and record.

Table S2: Retained examples and locations of their supporting records.

Example	What to inspect	Complete record location
Harbor	Source intervals versus query window (Figure S3)	Compact calls 1, 4–6
Orchard	Connected ownership slice, extra locations and attribution (Figures S1 and S2)	Compact calls 2, 7–9
Museum	Fresh synthetic comparison, including relevance errors	Compact calls 3, 10–12
Fable	Parallel actions, passive wording and conditional scope (Figure S4 and Section 6)	Prose calls 1, 4–5
Alice	Wrong reading participant and literal thought object (main paper)	Prose calls 2, 6–7
Holmes	Syntax failure and unresolved interpretation (main paper)	Prose calls 3, 8–9

The main Orchard figure is an illustrative selection from an existing evaluation, not a prospectively sampled new success. The views in Figures S1–S4 include extra irrelevant predictions, partial meanings and failures. No record was repaired for display, and no score was recomputed.

2. Orchard: correct relationships can still be irrelevant

Both views in Figure S1 preserve all five requested facts. The contextual output also returns seven locations. These are supported by the supplied story but outside this question. They remain visible and in the precision denominator. No belief records are present in this contextual ownership output. The panels share horizontal anchors and node categories, not an inferred identity mapping across different strings.

3. Orchard: attribution belongs to the assertion

Each object in Figure S2 connects to two places under different epistemic scopes. Copper Book’s South Loft location is attributed to Soren’s belief, while River Room is narrated reality. Ivory Vase similarly separates Jun’s belief from narration. Holder names remain edge metadata even though the names also occur elsewhere in the broader extraction. This selected graph has two disconnected components. No connector has been invented to make it appear more cohesive.

4. Harbor: preserve source duration, select by overlap

In Figure S3, the blue band is the requested window $[2, 4)$, not rewritten validity. Filled starts and open ends mark the full generated intervals. All four selected A location records coincide with four B records. The two carrying records in the figure are additional B predictions, not part of the location answer. All six B records remain scored. No relationship is given extra temporal precision because it was observed during the question window.

Source display: S5–S8 of the fourteen supplied sentences. Complete evidence and both outputs remain in the compact records. The main paper reports their unchanged strict scores in Table 1.

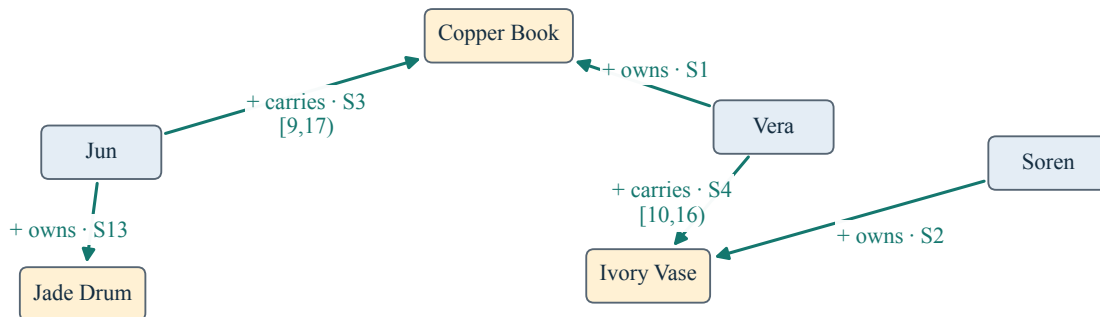
5. Fable: parallel actions and separate endpoint identities

All eight A and five B action records appear in Figure S4. A’s five nodes preserve its two different rope strings. B’s four nodes use a single literal ropes endpoint. This is not a corrected alias alignment. The complete source is the 133-word fable retained in the prose dataset. A’s ambiguous awakening and partial conditional plea remain in the graph.

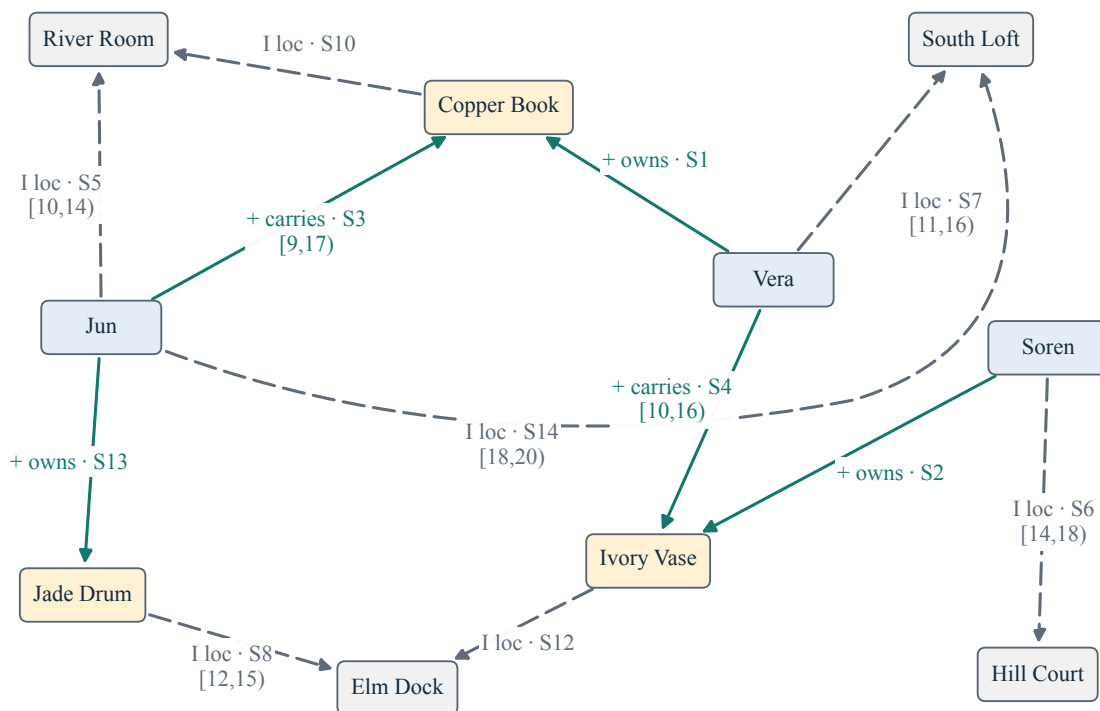
Exact executed ownership/carrying question: Who owns or carries what? Return all and only directly narrated ownership and carrying relationships, retaining every stated interval.

Orchard ownership · fixed selection versus contextual extraction

A: five selected records from call 2



B: independent contextual call 7 · all twelve records



+ supported and relevant. I supported but irrelevant (dashed). Colours assess records, not selection. Node fills are display categories. loc = located in. All holder/attitude values null. Unprinted bounds null.

Figure S1: Ownership and carrying networks from fixed selection (A) and contextual extraction (B).
A: 5/5 matched, 5 targets. P/R/F1 = 1.000/1.000/1.000. **B:** 5/12 matched, 5 targets. P/R/F1 = 0.417/1.000/0.588.

Exact executed question: Who explicitly believes what, and what reality is separately narrated about those same objects? Return each underlying believed relationship with its attribution and each matching narrated reality without attribution. Omit unrelated facts.

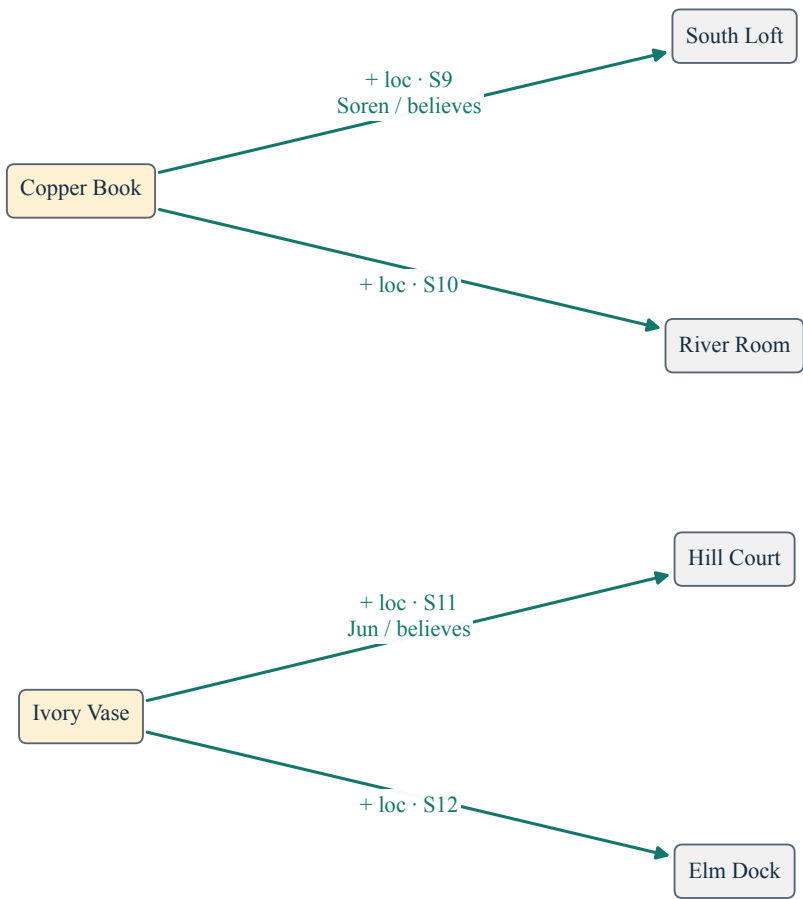
Orchard · believed and narrated locations

S9 Soren believes that Copper Book is in South Loft.

S10 In reality, Copper Book is in River Room.

S11 Jun believes that Ivory Vase is in Hill Court.

S12 In reality, Ivory Vase is in Elm Dock.



A: fixed selection from call 2. All four records on six endpoint nodes. Holder names are qualifications, not additional nodes or edges. All validity bounds null. + supported under the frozen evaluation. loc = located in.

Figure S2: Believed and narrated locations in Orchard’s fixed-selection network.
Fixed-selection score: 4/4 matched, 4 targets. P/R/F1 = 1.000/1.000/1.000.

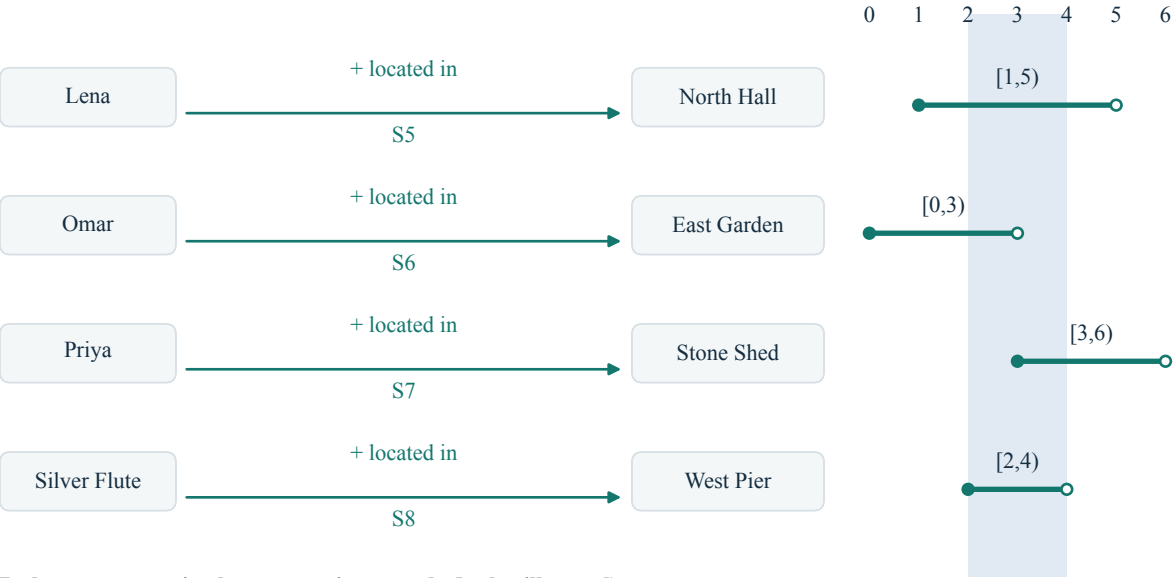
Exact executed question: Which directly narrated location relationships have an explicitly stated interval overlapping day 2 up to but not including day 4? Return all of them with their full evidence-stated intervals, not clipped to the question window. Exclude undated locations and beliefs.

Harbor · full intervals, not clipped to the question

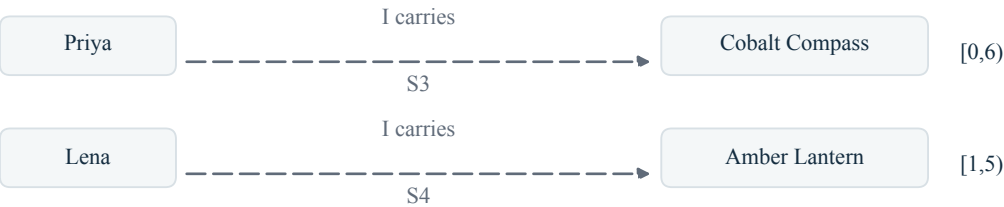
- S5 Lena is in North Hall from day 1 up to but not including day 5.
- S6 Omar is in East Garden from day 0 up to but not including day 3.
- S7 Priya is in Stone Shed from day 3 up to but not including day 6.
- S8 Silver Flute is in West Pier from day 2 up to but not including day 4.

Request: narrated locations overlapping [2,4), with full source intervals.

A and B: four identical location records, generated independently



B also returns two irrelevant carrying records (both still scored)



+ supported I irrelevant; all holder/attitude values null. Shading = query window; filled start / open end = source validity.

Figure S3: Full generated validity intervals and the Harbor location query window.

6. Interpreting agreement and preserved meaning

Two fable examples illustrate why strict agreement and semantic coverage are assessed separately.

Preserved meaning. The record Lion | caught_by | hunters | S4 correctly expresses the capture described in S4. It fails strict matching because the frozen reference alternatives do not include this passive wording. The semantic assessment therefore identifies useful information that the strict score does not credit.

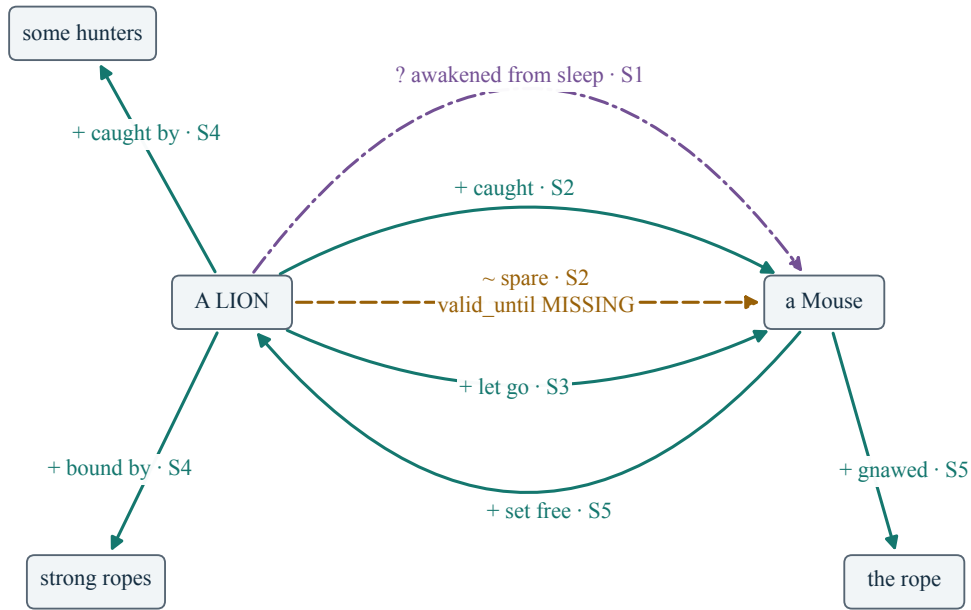
Partial meaning. The record Mouse | running_over | Lion | S1 captures the Mouse’s movement over the Lion, but S1 specifically describes movement over his face. The actor and broad action are preserved, while the more precise endpoint is omitted. This receives partial support rather than complete target coverage.

These examples demonstrate the value of inspecting network relationships alongside their source passages. Readers can distinguish alternative wording from missing detail while examining the original outputs and unchanged scores.

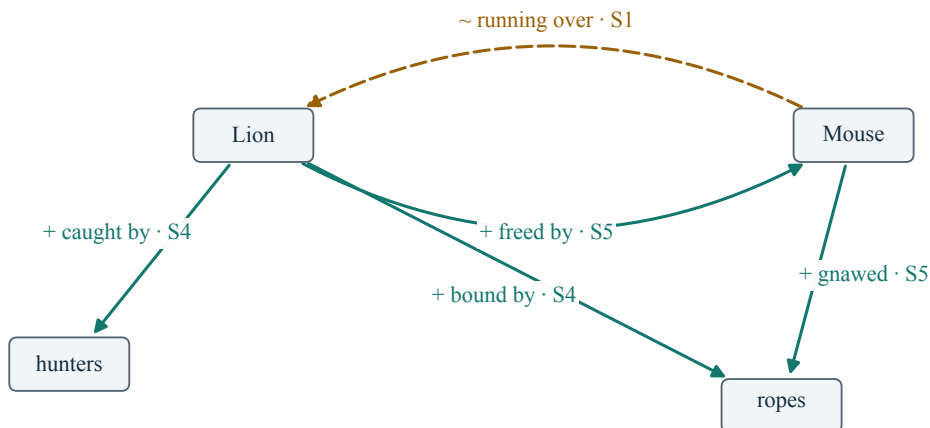
Same executed question as main-paper Figure 2: Which physical running/waking, capture, binding, rope-gnawing and release relationships involve the Lion, Mouse, hunters and ropes? Exclude dialogue, intentions, laughter and the moral.

Fable actions · full networks with literal endpoint identities

A: fixed selection · all eight records



B: independent contextual extraction · all five records



+ supported meaning. ~ partial. ? unresolved. Exact endpoint strings remain separate: A has “strong ropes” and “the rope”, whereas B uses “ropes”. All attribution and bounds null except A’s missing valid_until field on spare (literal extra key “valid until”).

Figure S4: Complete fable action networks from fixed selection (A) and contextual extraction (B).
A: P/R/F1 = 0.625/0.625/0.625. **B:** P/R/F1 = 0.400/0.250/0.308.